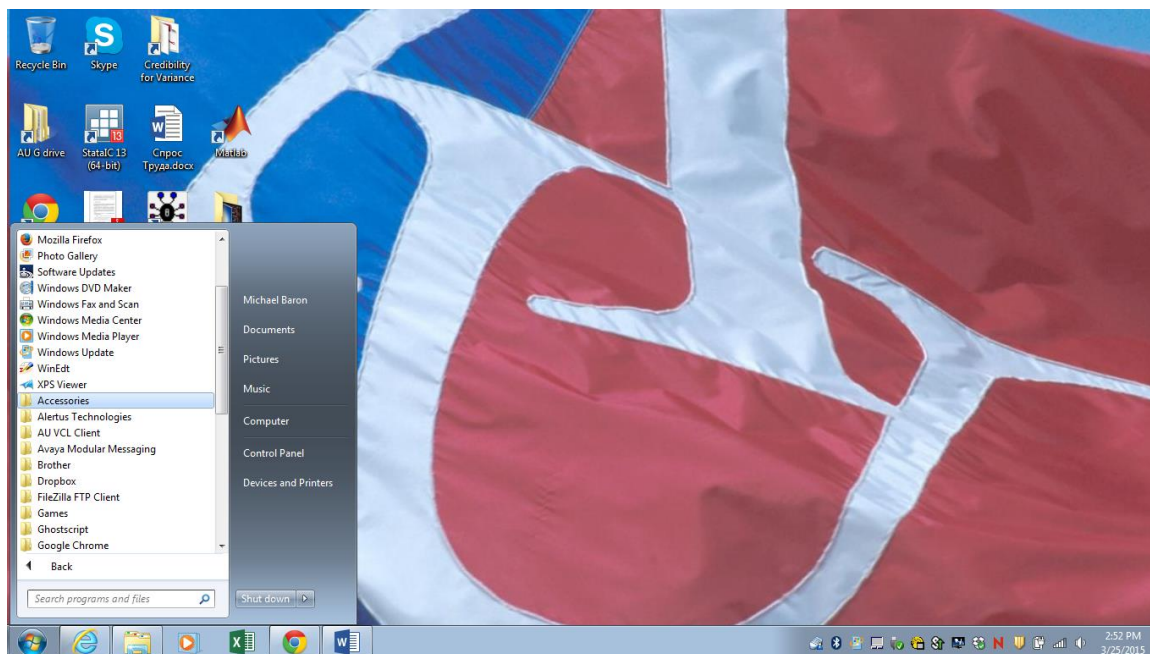
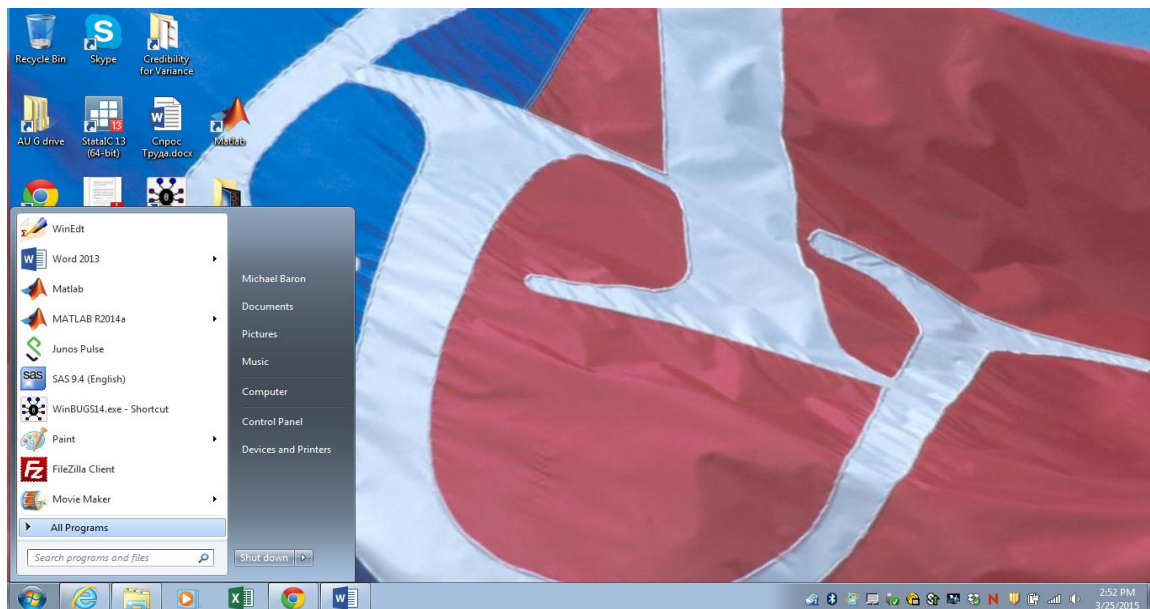


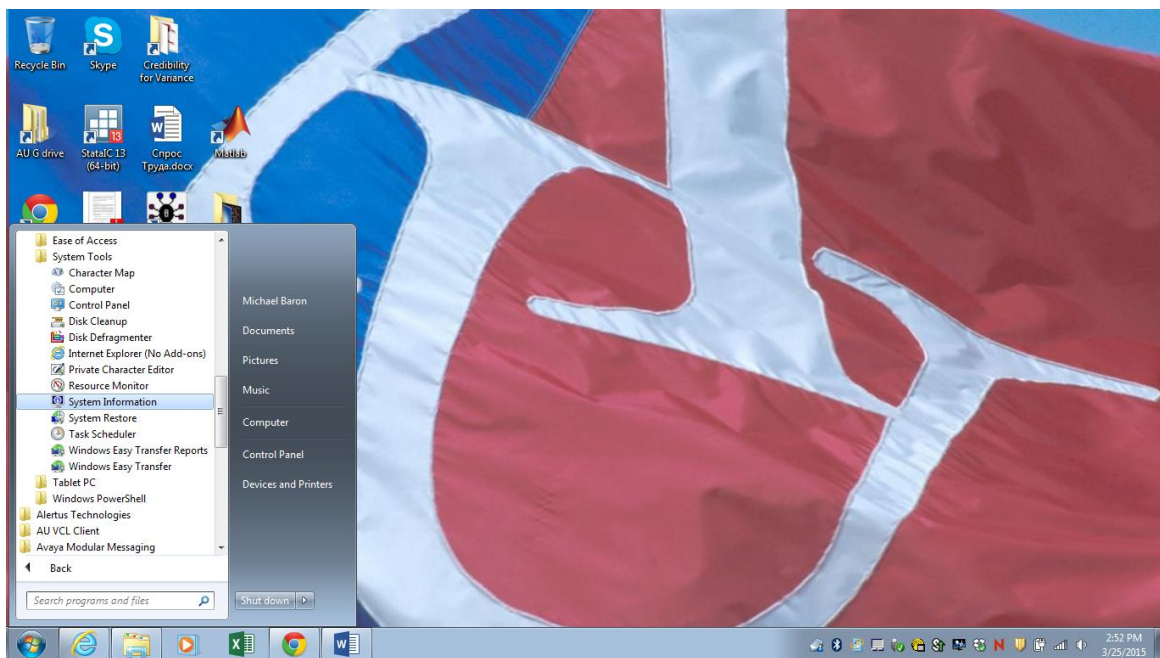
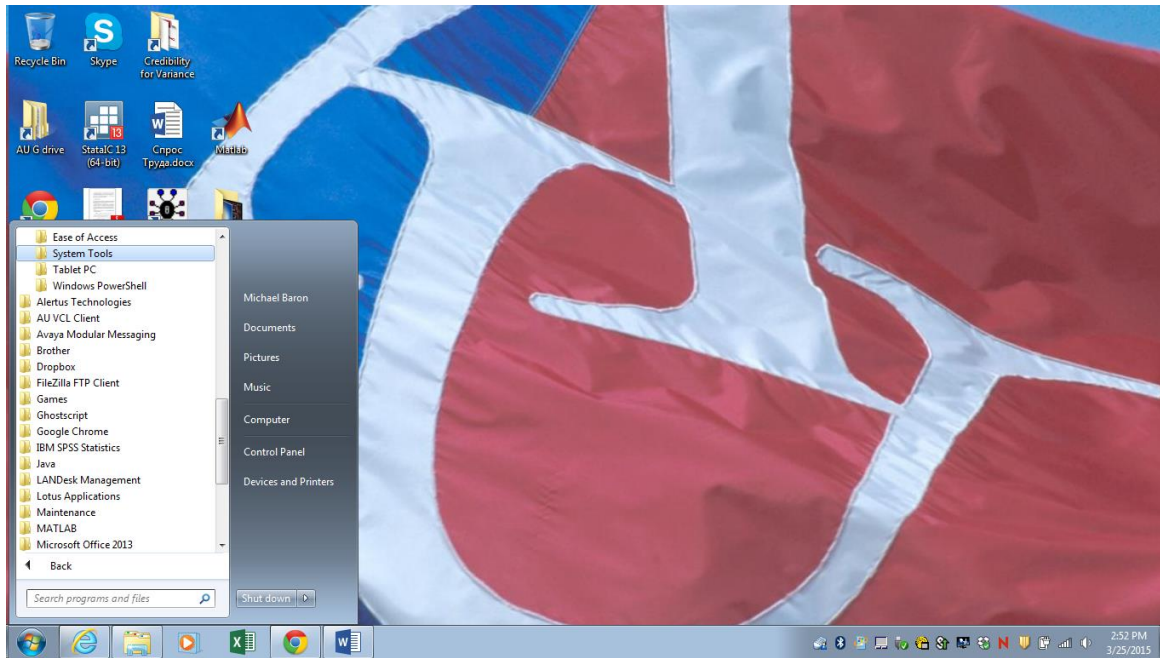
BUGS Software (Bayesian Inference Using Gibbs Sampling)

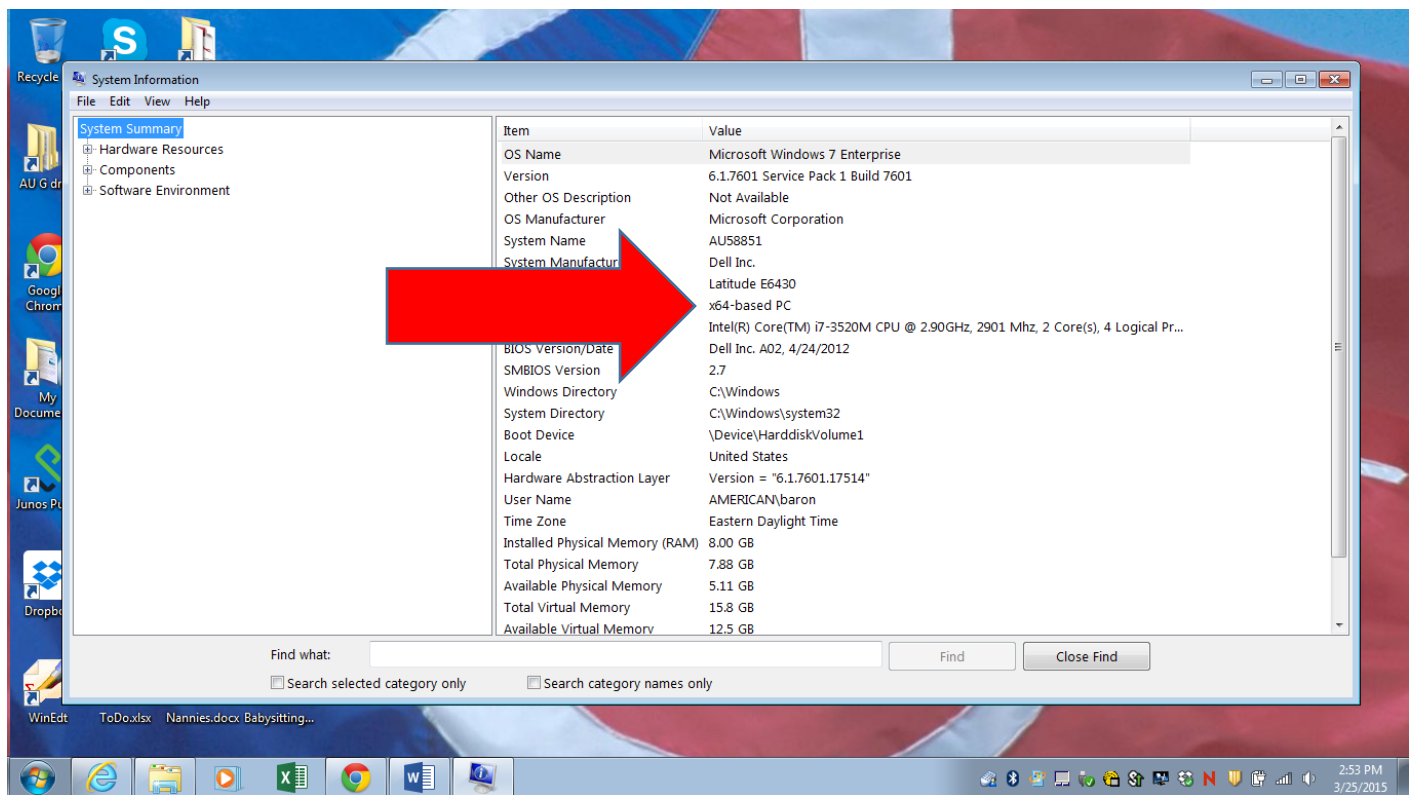
1. Installation

Step 1. Preparation. Find out whether you are running a 32-bit or a 64-bit machine. The answer is in “System information”:

Go to Start → Programs → Accessories → System Tools → System Information

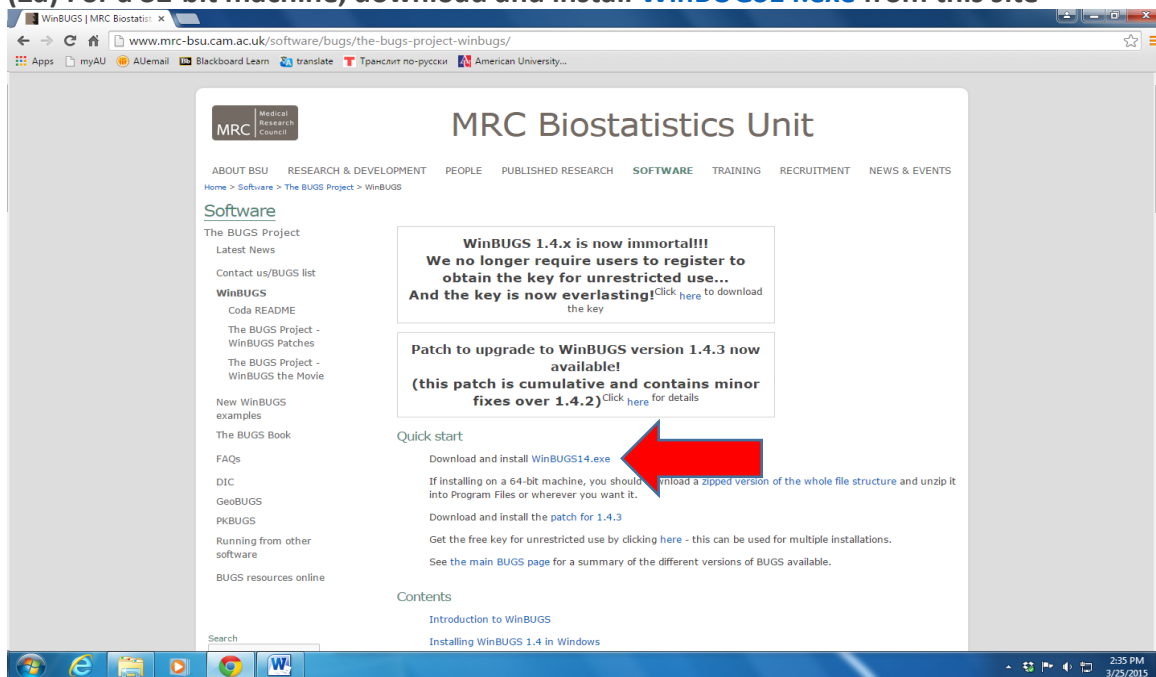






Step 2. WinBUGS, now a free version of BUGS, is available from <http://www.mrc-bsu.cam.ac.uk/software/bugs/the-bugs-project-winbugs/>

(2a) For a 32-bit machine, download and install **WinBUGS14.exe** from this site



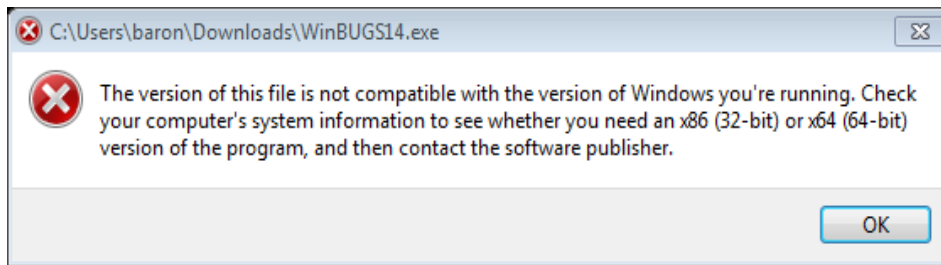
(2b) For a 64-bit machine, download a **zipped version** ...

The screenshot shows the MRC Biostatistics Unit website. The main heading is "MRC Biostatistics Unit". Below it, there is a navigation menu with links: ABOUT BSU, RESEARCH & DEVELOPMENT, PEOPLE, PUBLISHED RESEARCH, SOFTWARE, TRAINING, RECRUITMENT, NEWS & EVENTS. The "Software" section is highlighted. On the left, there is a sidebar with links: The BUGS Project, Latest News, Contact us/BUGS list, WinBUGS, Coda README, The BUGS Project - WinBUGS Patches, The BUGS Project - WinBUGS the Movie, New WinBUGS examples, The BUGS Book, FAQs, DIC, GeoBUGS, PKBUGS, Running from other software, and BUGS resources online. The main content area has a large announcement: "WinBUGS 1.4.x is now immortal!!! We no longer require users to register to obtain the key for unrestricted use... And the key is now everlasting! Click here to download the key". Below this, there is a section for "Patch to upgrade to WinBUGS version 1.4.3 now available! (this patch is cumulative and contains minor fixes over 1.4.2) Click here for details". The "Quick start" section contains instructions: "Download and install WinBUGS14.exe", "If installing on a 64-bit machine, you should download a zipped version of the whole file structure and unzip it into Program Files or wherever you want it.", "Download and install the patch for 1.4.3", "Get the free key for unrestricted use by clicking here - this can be used for multiple installations.", and "See the main BUGS page for a summary of the different versions of BUGS available.". A purple arrow points to the "zipped version" link in the "Quick start" section. The "Contents" section lists "Introduction to WinBUGS" and "Installing WinBUGS 1.4 in Windows".

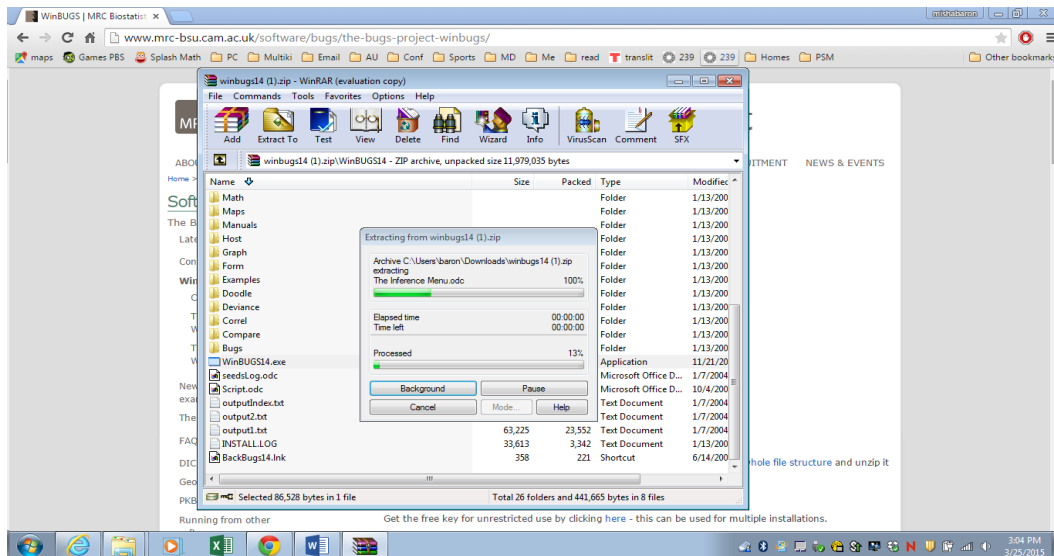
... and unzip it.

The screenshot shows a Windows file explorer window titled "winbugs14 (1).zip - WinRAR (evaluation copy)". The window displays the contents of the "winbugs14 (1).zip" file, which is a ZIP archive, unpacked size 11,979,035 bytes. The file list shows various folders and files, including "Math", "Maps", "Manuals", "Host", "Graph", "Form", "Examples", "Doodle", "Deviance", "Correl", "Compare", "Bugs", "WinBUGS14.exe", "seedLog.odc", "Script.odc", "outputIndex.txt", "output2.txt", "output1.txt", "INSTALLLOG", and "BackBugs14.lnk". The "WinBUGS14.exe" file is highlighted. The status bar at the bottom indicates "Selected 86,528 bytes in 1 file" and "Total 26 folders and 441,665 bytes in 8 files".

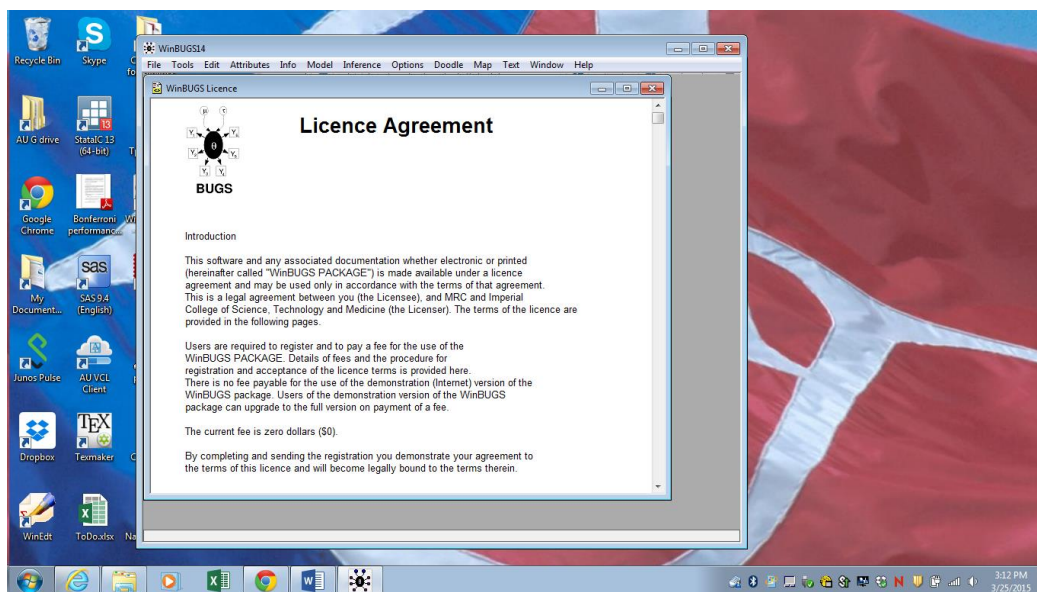
If you are not sure whether you have a 32-bit or 64-bit computer, and you try a wrong version, the system will complain:



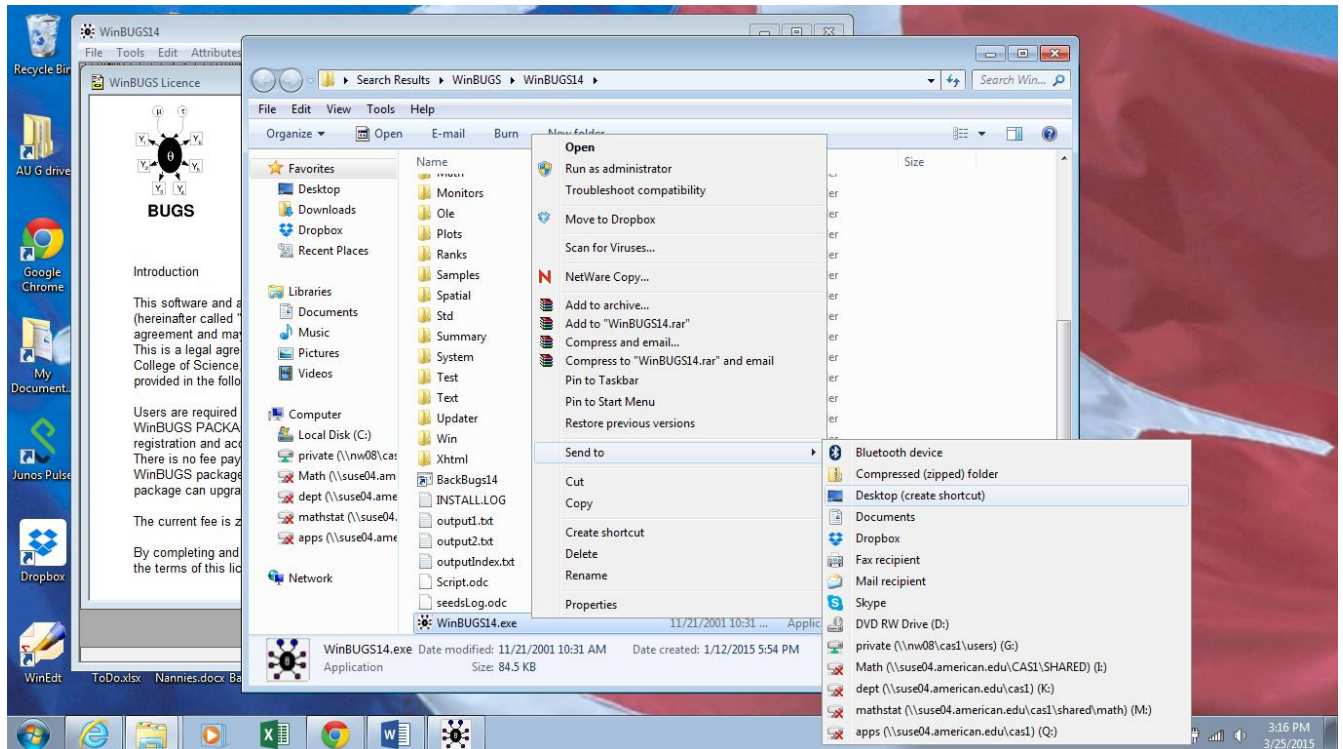
Execute the file WinBUGS14.exe



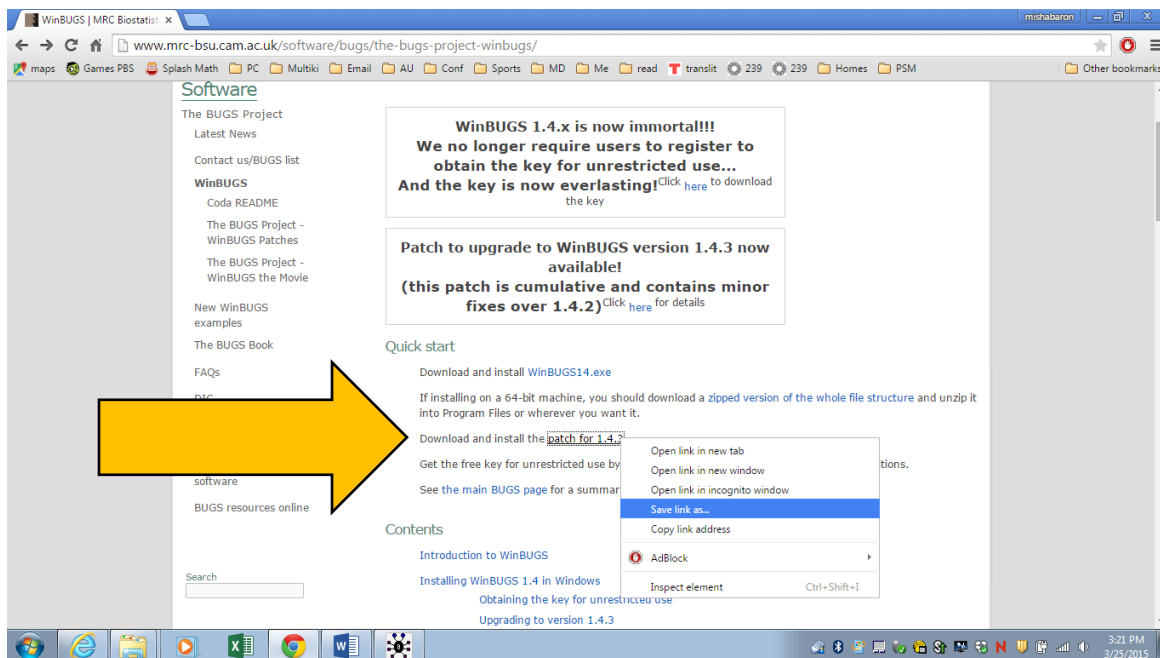
WinBUGS opens:



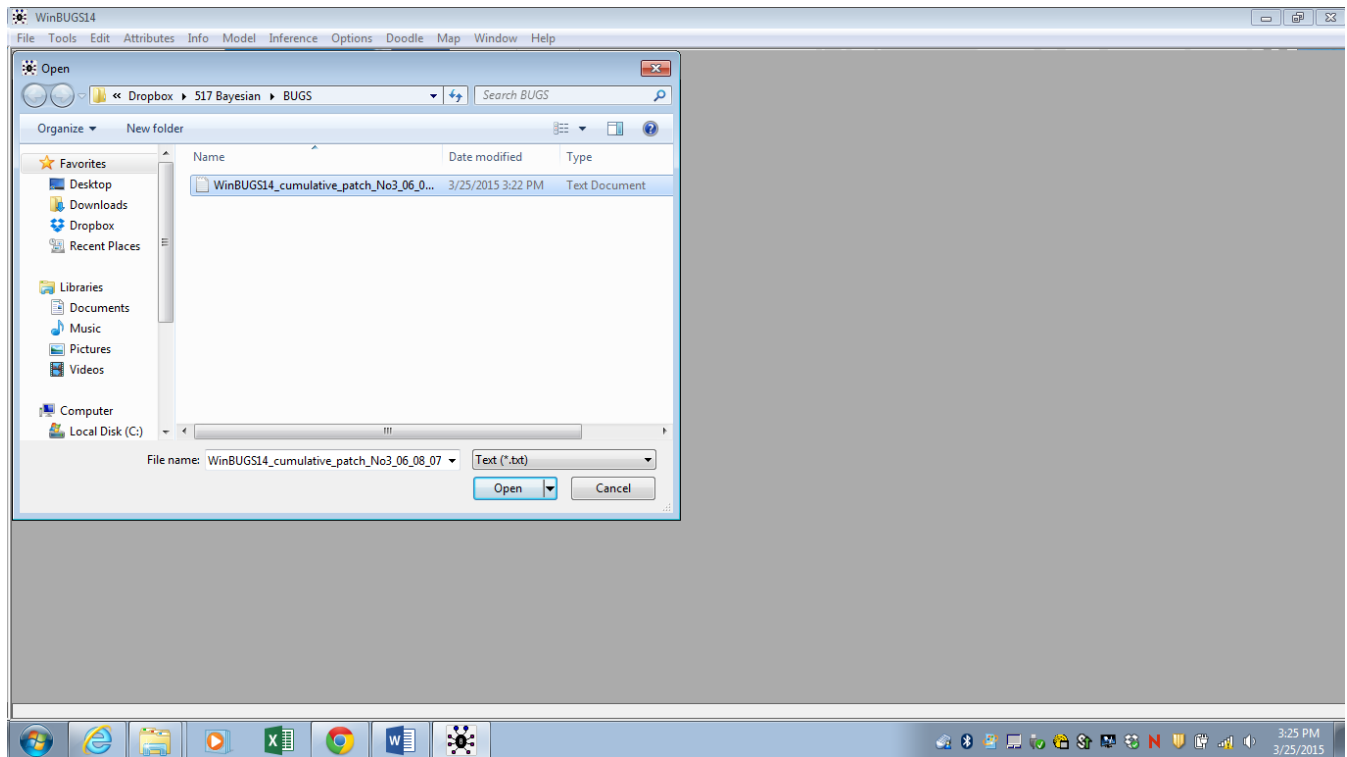
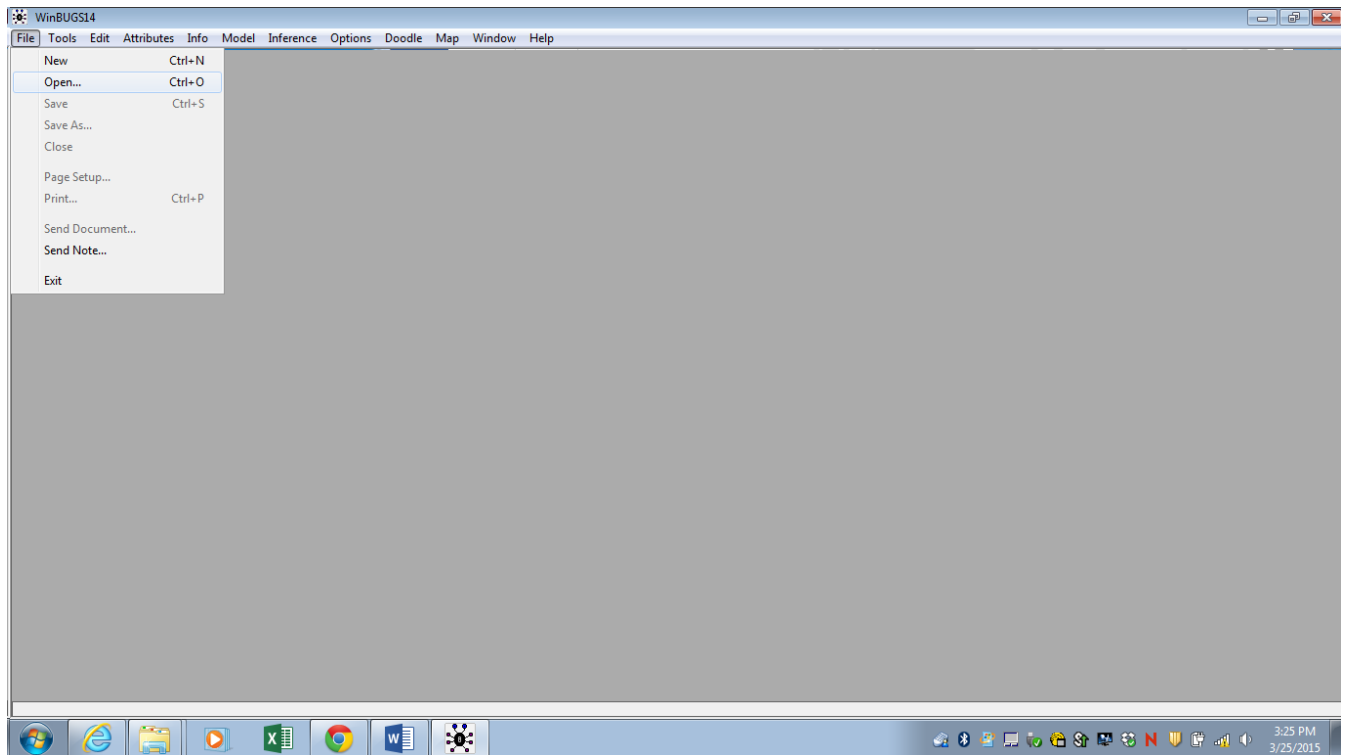
Create a shortcut of WinBUGS.exe to desktop for your convenience by sending the file to desktop (create shortcut).

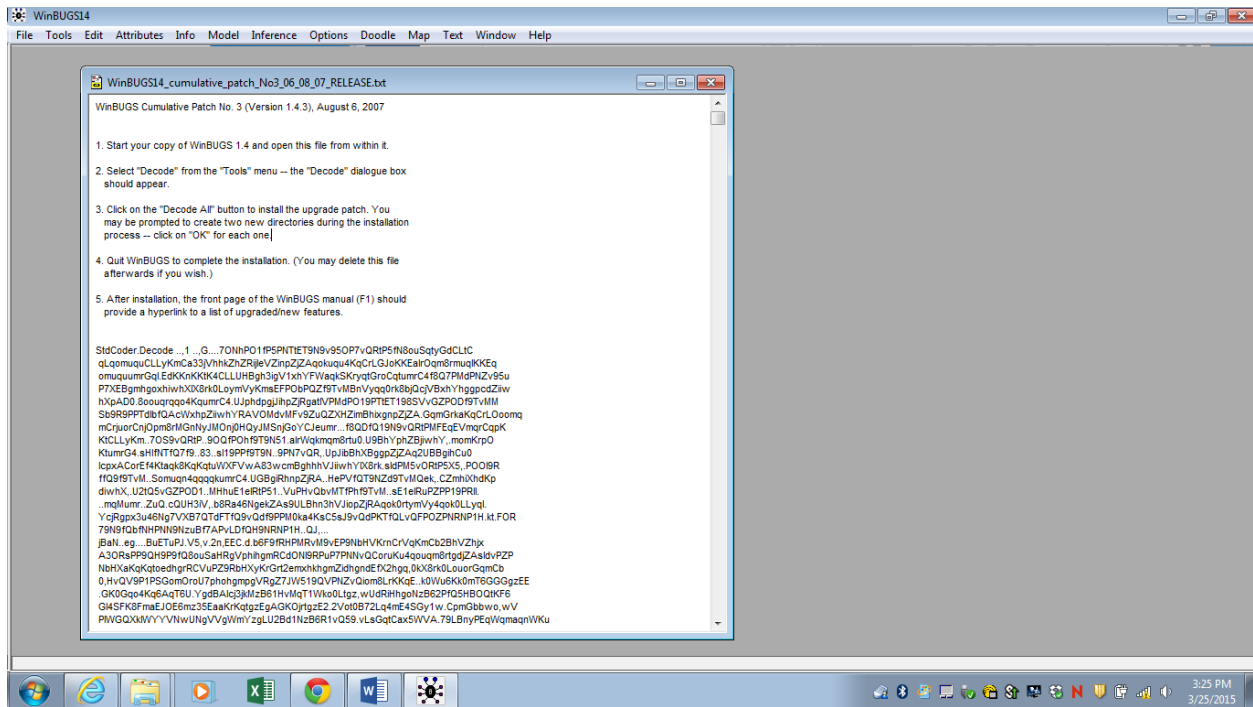


Step 3. Registration for eternal free use. On the same site, find “Download and install the patch for 1.4.3”, right-click on it, save the txt file, then open it in WinBUGS.

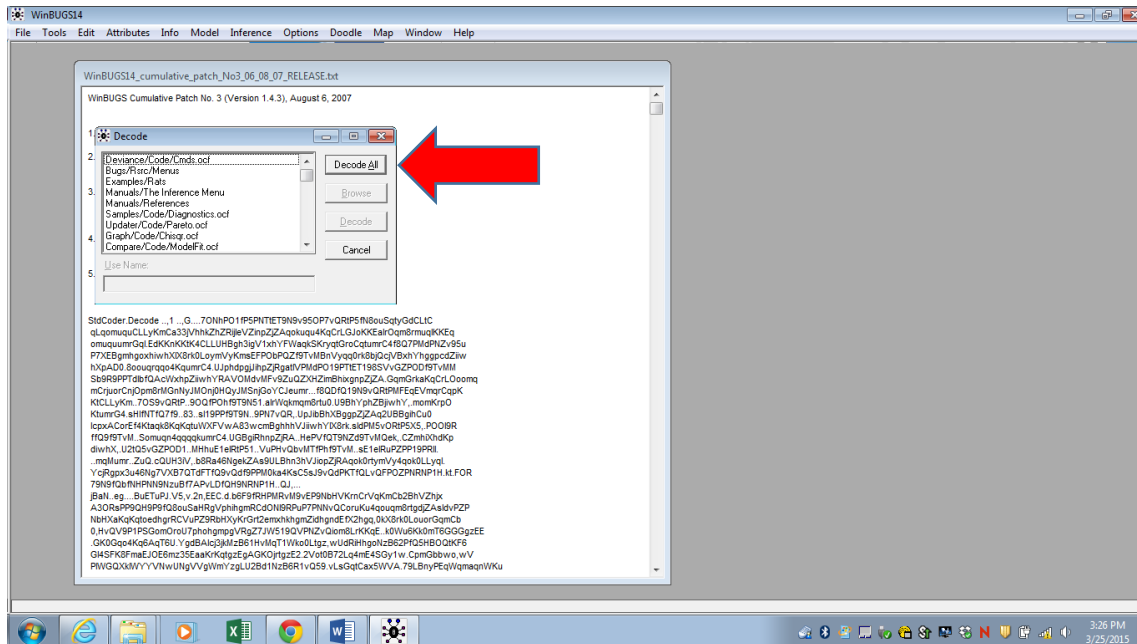


In WinBUGS, go to File → Open, and open this text file with the patch.

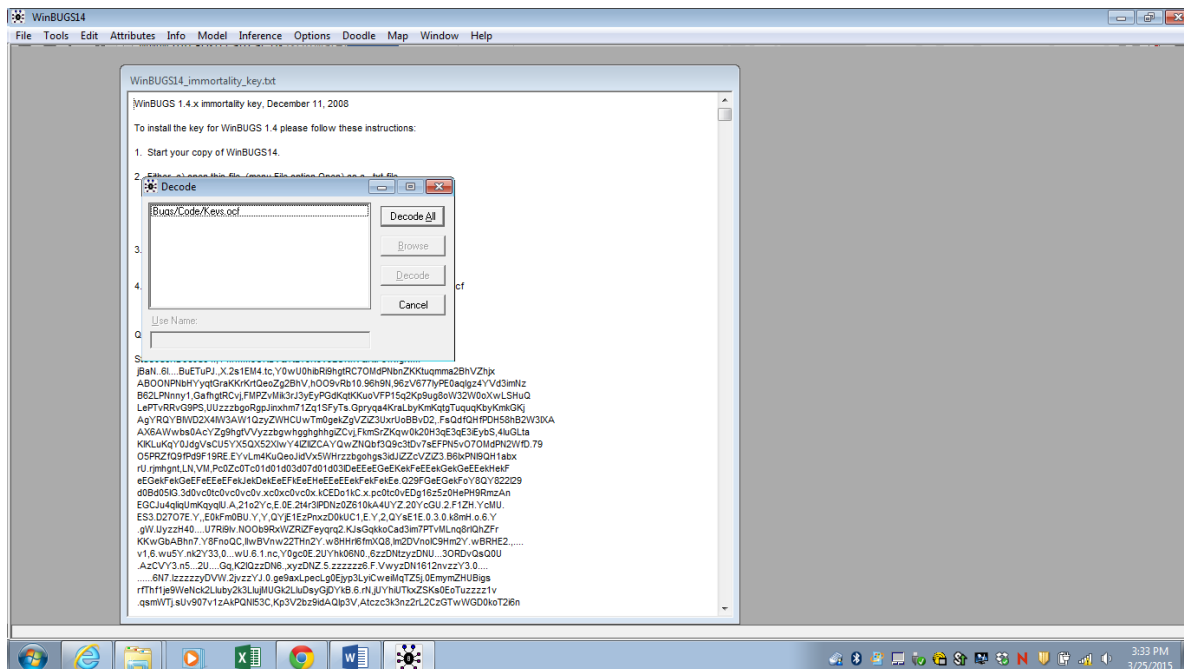
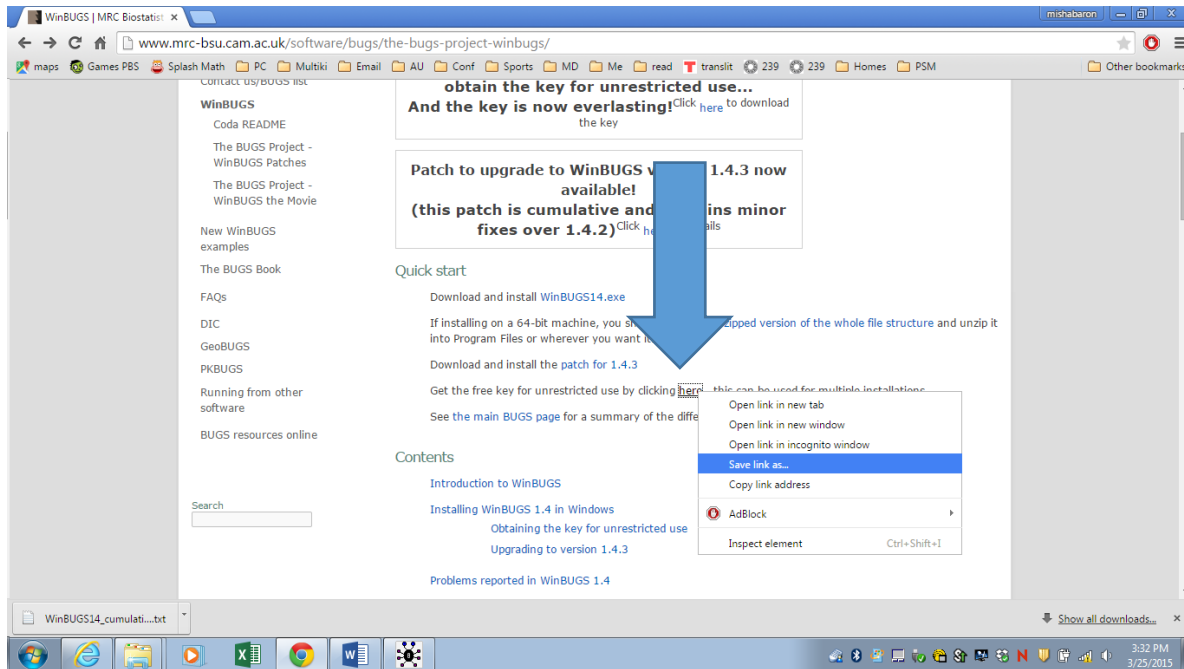




Go to Tools → Decode, then press “Decode all”.



From the same site, get the “immortality key” for unrestricted use and do the same with it - save as txt file, open it in WinBUGS, Decode, and choose “Decode all”.



2. Basic Steps

Write a program

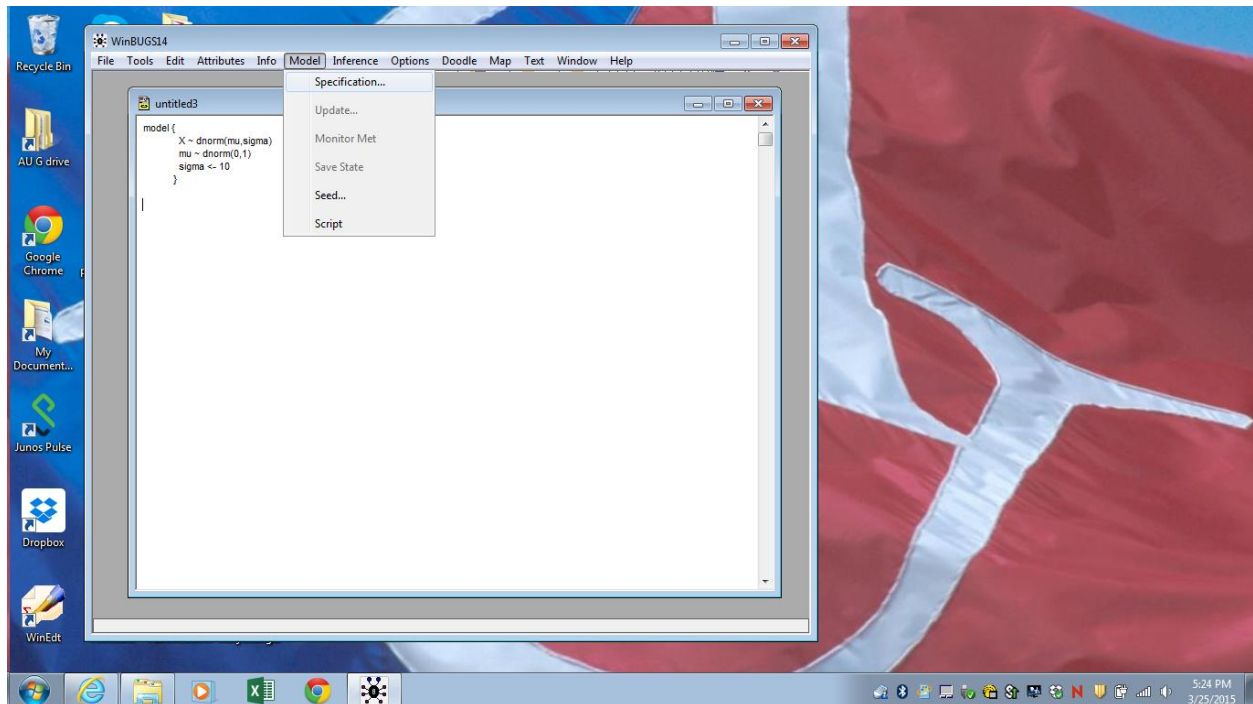
```
model {  
  X ~ dnorm(mu,sigma)  
  mu ~ dnorm(0,1)  
  sigma <- 10  
}
```

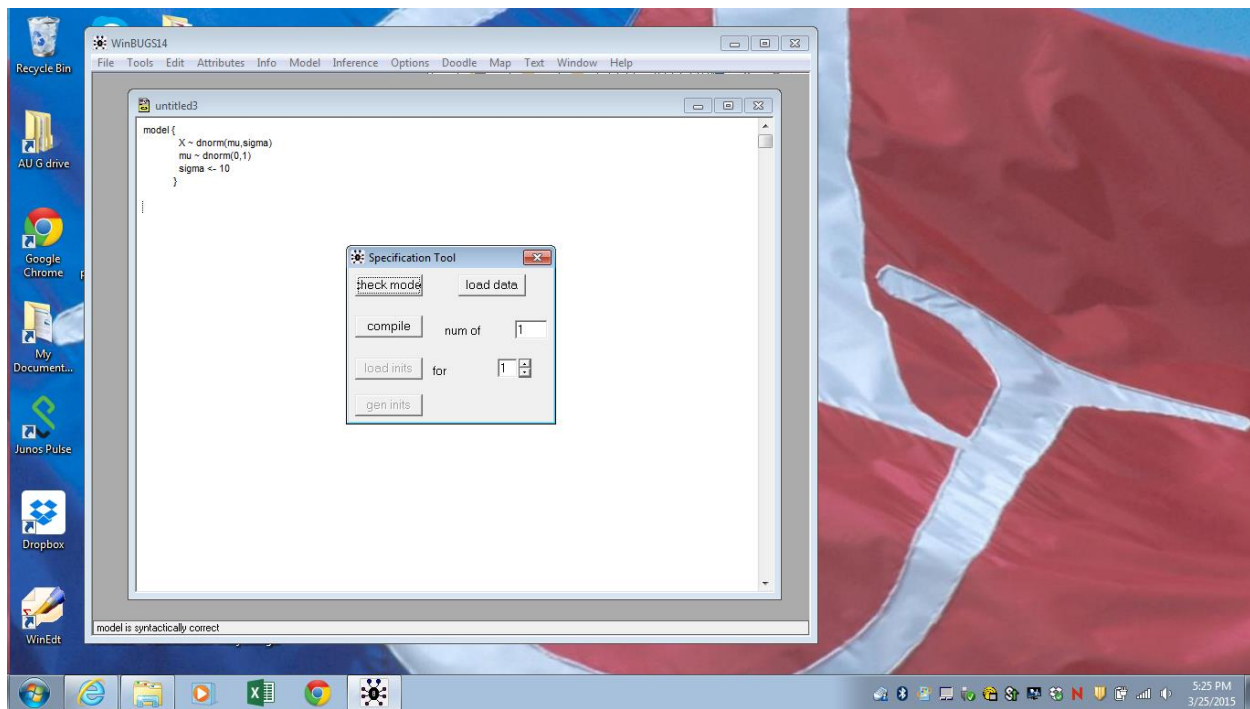
Verify it and compile:

Model -> specification tool -> check model

Response:

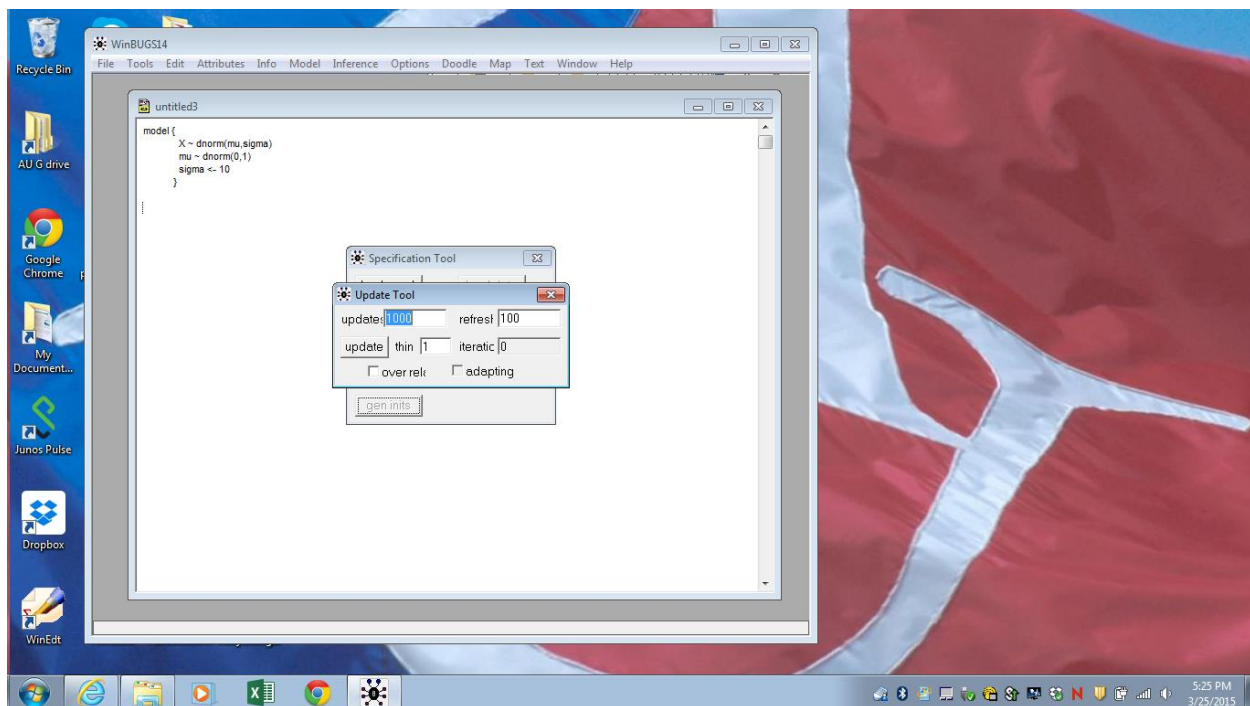
synthetically correct





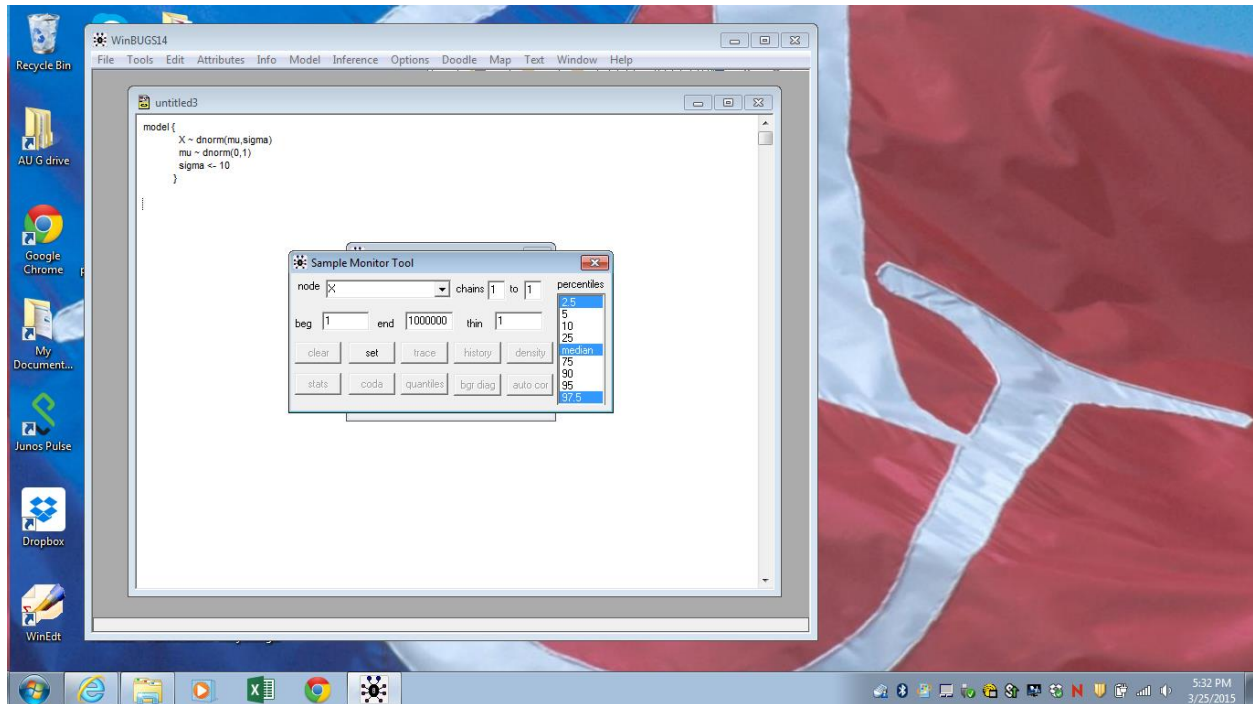
→ Compile Response: compiled
 → Gen inits Response: model initialized

Model -> update



Specify variables we want to monitor.

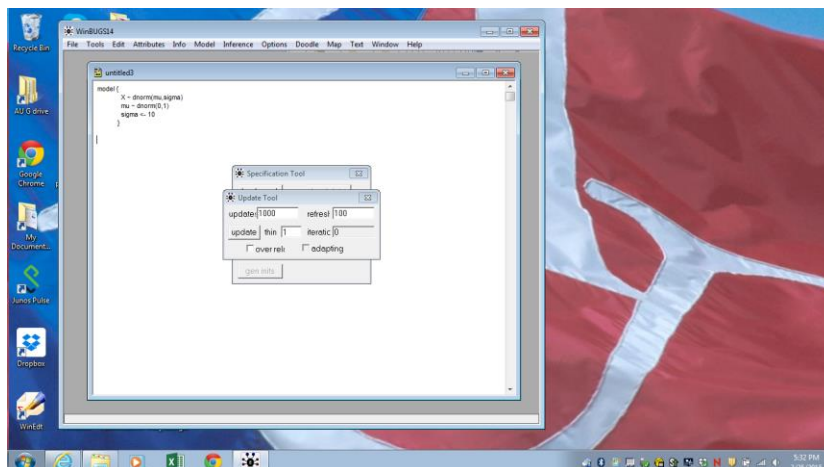
Inference -> Samples



- ➔ node -> X -> set
- ➔ node -> mu -> set
- ➔
- ➔ node -> * -> trace

Run the simulations.

Update tool window -> update



Collect the desired statistics and observe the desired graphs for the monitored variables

Sample Monitor window -> stats

Sample Monitor window -> trace

Etc.

